

Geometric Spanning Trees that Minimizes the Wiener Index

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This week

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- Read through and understand the weak NP-Hardness reduction.

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- Read through and understand the weak NP-Hardness reduction.
- Think of suitable gadgets to prove the unbounded version is also NP-Hard.

EUCLIDEAN WIENER INDEX TREE PROBLEM (EWITP)

INSTANCE: A set P of n points in $\mathbb{R}^2$, a cost W , and a budget B .

QUESTION: Does there exist a spanning tree T of P such that $W(T) \leq W$ and $wt(T) \leq B$?

Here, $W(T)$ denotes the Wiener index of the tree T and $wt(T)$ denotes the total weight of the tree T .

PARTITION

INSTANCE: A set $X = \{x_1, x_2, \dots, x_n\}$, where $x_i \in \mathbb{Z}$ such that

$$R = \sum_{i=1}^n x_i.$$

QUESTION: Does there exists a subset $S \subseteq X$ such that $\sum_{x_i \in S} x_i = \frac{R}{2}$.

The following reduction is from [Abu-Affash et al., 2023].

Definition

Let P be a set of n points in $\mathbb{R}^2$ and let T be a spanning tree of P . For an edge $(p, q) \in T$, we define $N_T(p, q) = |\{x \in P : \delta_T(p, x) < \delta_T(q, x)\}|$ be the number of points in T that are closer to p than to q .

Hardness Proof

Given an instance $X = \{x_1, x_2, \dots, x_n\}$ of the **PARTITION** instance where $x_i \in \mathbb{Z}$:

- Construct a set P of $m = n^3 + 3n + 1$ points.
 - A point s and n points $p_1, p_2, \dots, p_n$ located equally spaced on a circle of radius nR centered at s .
 - A cluster C of n^3 points located at the center of the circle of radius $\frac{1}{n^{10}R}$ centered at S .
 - For each $1 \leq i \leq n$, we have two points l_i and r_i both of distance x_i from p_i and the distance between them is $\frac{x_i}{2}$.
- Set

$$B = \left(n^2 + \frac{7}{4}\right)R + \frac{1}{n^7R}$$

$$\begin{aligned} W &= 3n^2(m-3)R + \left(\frac{9}{4}m - \frac{13}{4}\right)R + \frac{1}{n^4R} + \frac{3}{n^6R} \\ &= 3n^5R + \frac{45}{4}n^3R - 6n^2R + \frac{27}{4}nR - R + \frac{1}{n^4R} + \frac{3}{n^6R} \end{aligned}$$

Hardness Proof

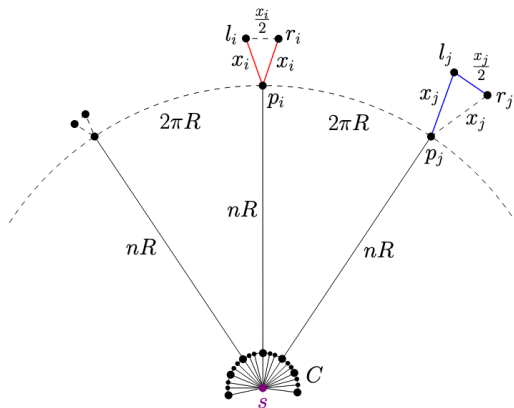


Figure 1: The set P produced by the reduction. Connecting the points l_j, r_j , and p_j for $x_j \in S$ (Blue) and connecting the points l_i, r_i , and p_i for $x_i \in X \setminus S$ (Red).

Assume there exists a set $S \subseteq X$ such that $\sum_{x_i \in S}^{x_i} = \frac{R}{2}$.

Construct a spanning tree T for the points of P as follows:

- $E_1 = \{(s, p_i) : 1 \leq i \leq n\}$
- $E_2 = \{(p_i, l_j), (p_i, r_i) : x_i \in S\}$
- $E_3 = \{(p_j, l_j), (l_j, r_j) : x_j \in X \setminus S\}$, and
- $E_4 = \{(s, p) : p \in C\}$

$$\begin{aligned}wt(T) &= wt(E_1) + wt(E_2) + wt(E_3) + wt(E_4) \\&= n^2R + R + \frac{3}{4}R + n^3 \cdot \frac{1}{n^{10}R} \\&= \left(n^2 + \frac{7}{4}\right)R + \frac{1}{n^7R} \\&= B\end{aligned}$$

Wiener Index of the Tree

$$\begin{aligned} W(T) &= \sum_{(p,q) \in T} N_T(p,q) \cdot N_T(q,p) \cdot |pq| \\ &= \sum_{(p,q) \in E_1} N_T(p,q) \cdot N_T(q,p) \cdot |pq| \\ &\quad + \sum_{(p,q) \in E_2} N_T(p,q) \cdot N_T(q,p) \cdot |pq| \\ &\quad + \sum_{(p,q) \in E_3} N_T(p,q) \cdot N_T(q,p) \cdot |pq| \\ &\quad + \sum_{(p,q) \in E_4} N_T(p,q) \cdot N_T(q,p) \cdot |pq| \\ &= W(\text{Chalk \& Talk}) \end{aligned}$$

It is a matter of computation to show that:

Theorem (Claim 4)

There are exactly n edges in T between the points in $C \cup \{s\}$ and the others.

The paper shows it considering the cases of there are more than n edges and less than n edges and shows a contradiction.

Let P_i denote the triplet $\{p_i, l_i, r_i\}$ for $1 \leq i \leq n$.

- From Claim 4, it follows that for every $1 \leq i \leq n$, there is exactly one edge (p, q) in T for which $p \in C \cup \{s\}$ and $q \in P_i$. Note that $q = p_i$.
- Thus, in every P_i , we have $(p_i, l_i) \in T$ or $(p_i, r_i) \in T$.
- WLOG let $(p_i, l_i) \in T$:
 - Either $(p_i, r_i) \in T$ OR $(l_i, r_i) \in T$.
 - Let $S \subseteq X$ such that $x_i \in S \iff (p_i, r_i) \in T$.
 - Let $R' = \sum_{x_i \in S} x_i$.

They then argue that if $R' \neq \frac{R}{2}$, then either $wt(T) > B$ or $W(T) > W$.

This concludes the hardness result.

Why is the bound $wt(T) \leq B$ important?

The weight bound is what forces the gadgets to connect in a very specific way (open or closed), which in turn encodes the binary choices in **PARTITION**.

- n gadget for each integer $x_i \in X$, giving two connection options:
 - **Open:** (p_i, l_i) and (p_i, r_i) contributing $2x_i$ weight.
 - **Closed:** (p_i, l_i) and (l_i, r_i) contributing $\frac{3x_i}{2}$ weight.

If we takeout the weight bound:

- We could just have everything open, which might reduce the Wiener index (Since open gadgets keep gadget paths short), but increase the weight
- In the unbounded case, we would have no reason to avoid such trees.

Next steps?

- Preliminarily, I think the modification to the reduction needs to happen such that the penalty is no longer on the weight but solely on the Wiener Index. Of course, this is harder to prove.
- Primarily, construct gadgets so that the only correct combination of open/closed configurations minimize the Wiener Index.

Next steps?

Construct spanning tree T of point set P and show:

- If **PARTITION** is solvable, then $W(T) \leq W$.
- Otherwise, ALL trees have Wiener index $> W + \delta$ for some $\delta > 0$.

Next week tasks!

- CCCG/WADS (Yay)! Read the pre-prints on arxiv and see which ones I think I would enjoy listening in the most.
- Keep thinking about the gadgets that we could use.



Abu-Affash, A. K., Carmi, P., Luwisch, O., and Mitchell, J. S. B. (2023).

Geometric spanning trees minimizing the wiener index.

The end.
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